

Independent Validation: pH Prediction from TCMC Posterior

A matrix (Phase 2, 10,000-p NUTS) $\rightarrow$ Hamilton ODE $\rightarrow$ $\text{pH} = \beta_0 + \beta_{\text{So}}\phi_{\text{So}} + \beta_{\text{An}}\phi_{\text{An}} + \beta_{\text{Vei}}\phi_{\text{Vei}} + \beta_{\text{Fn}}\phi_{\text{Fn}} + \beta_{\text{Pg}}\phi_{\text{Pg}}$ (β from experimental LOO-CV, $N=90$; pH was not used in TCMC calibration)

